Forecasting Hourly Visitor Arrivals at Liangzhu Archaeological Ruins Park

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Abstract

Accurate short-term visitor forecasting can help tourist attractions allocate staff, manage congestion, and improve visitor safety. This project investigates hourly entrance count at Liangzhu Archaeological Ruins Park in Hangzhou, China. Specifically, we download the raw entrance records from Hangzhou Data Open Platform. The records from June 2019 to June 2026 are aggregated by hour. Calendar and weather variables (including hour of day, day of week, public-holiday status, and weather condition) are incorporated as covariates (i.e, variables that can influence entrance count) . We complete missing records of early morning (1:00-6:00) with zeros. Only windows with entrance counts for 48 consecutive records are retained as samples. We use a dual Gated Recurrent Unit (GRU) neural network to build a forecasting model that uses the previous 24-hour entrance counts and the next 24-hour covariates to predict the next 24-hour entrance counts. The model achieves an MAE of 65.40, an RMSE of 157.79, and an SMAPE of 97.39%.

Keywords: visitor-arrival forecasting; time series; GRU; data preprocessing; heritage-site management

1. INTRODUCTION

Liangzhu Archaeological Ruins Park is located in Yuhang District, Hangzhou, Zhejiang Province. The park is recognized as a UNESCO World Heritage Site[1] and provides evidence of an early regional state based on rice agriculture in the lower Yangtze River during the Late Neolithic period. Today, the park serves as a protected cultural heritage site and a major cultural attraction. Figure 1 presents an aerial view of the park.

The entrance count of a cultural attraction changes substantially across different calendar and weather conditions. Reliable short-term forecasting can therefore support practical decisions such as staff scheduling, transport coordination, crowd control, and safety planning.

This project develops an hourly forecasting model for Liangzhu Archaeological Ruins Park. Its main work is summarized as follows:

1. **Data collection.** Entrance records are obtained from Hangzhou Data Open Platform. Weather data are collected from Open-Meteo API, while public holiday information is derived from a public Chinese holiday dataset.
2. **Data preprocessing and analysis.** Entrance records are aggregated by hour. Missing records between 1:00 and 6:00 are completed with zero entrance counts, and only windows with values for 48 consecutive records are retained as samples for modeling. Calendar and weather variables are also processed at an hourly resolution.
3. **Forecasting model development and evaluation.** A dual-GRU model is developed to predict entrance counts for the next 24 hours. One GRU encodes the entrance counts from the previous 24 hours, while the other encodes calendar and weather covariates from the next 24-hour period. Their final hidden states are combined and passed to a predictor that produces 24 hourly forecasts. Performance is evaluated on a chronologically separated test set using multiple error metrics.



Figure 1. Aerial view of Liangzhu Archaeological Ruins Park[2].

2. DATA PROCESSING AND EXPLORATORY ANALYSIS

2.1 Data Source

We download the raw entrance records from Hangzhou Data Open Platform[3]. The original records contain entrance counts. The available data cover the period from June 2019 to June 2026. Hourly weather condition and public holiday information are collected from Open-Meteo Historical Weather API[4] and the open-source project NateScarlet/holidaycn[5], respectively.

[1] https://whc.unesco.org/en/list/1592/

[2] https://whc.unesco.org/en/list/1592/gallery

[3] https://data.hangzhou.gov.cn

[4] https://archive-api.open-meteo.com/v1/archive

[5] https://www.jsdelivr.com/package/gh/NateScarlet/holiday-cn

Table 1. Summary of the processed dataset.

Characteristic	Value
Period	Jun. 2019–Jun. 2026
Number of original records	53,158
Number of records after completion	58,876
Number of samples	48,124

2.2 Hourly Aggregation

The raw data do not always contain exactly one record for each timestamp. In some cases, multiple entrance records are reported within the same timestamp. However, the dataset requires a single entrance count at each timestamp so that input and output sequences can be generated. Therefore, hourly aggregation is performed before sample generation.

First, the original timestamps are converted to a standard datetime format and truncated to the beginning of the corresponding hour. For example, records collected at 10:15 and 10:42 are both assigned to 10:00. The entrance counts of all records assigned to the same hour are then summed to obtain one entrance count. Thus, if the two records contain 18 and 27 entrance counts, respectively, the aggregated entrance count at 10:00 will be 45. Entrance counts are converted to a numeric format. Finally, the aggregated records are sorted chronologically, generating a time series with at most one record for each timestamp.

2.3 Missing Records Completion and Sample Generation

We identify some missing records within the time series (i.e, some timestamps do not have corresponding records) . These missing records require special treatment, as each sample has to contain 48 consecutive records. Without completing missing records, only 6,083 samples can be generated. This will discard a large proportion of the available entrance records and leave relatively few samples for model training. Completing missing records with zero is inappropriate, as the absence of a record does not necessarily mean that this timestamp has no entrance count. In particular, missing records during daytime opening hours may correspond to unreported positive entrance counts.

An exploratory inspection shows that a large proportion of missing records occurs between 1:00 and 6:00. Since these hours are outside the park's normal opening period, the entrance counts of these hours are assumed to be zero. Therefore, only missing records from 1:00 to 6:00 are completed with zero entrance counts. In total, 5,718 early-morning records are added, increasing the number of records from 53,158 to 58,876.

Finally, a sliding window with a length of 48 timestamps and a stride of 1 timestamp is applied to the time series to obtain candidate samples. A sample without missing records is retained. Each sample is then divided into two inputs: the entrance counts from the previous 24 hours and the calendar and weather covariates from the next 24-hour period. The entrance counts from the next 24 hours is used as the prediction target. This process generates 48,124 samples. Table 1 shows the summery of the processed dataset.

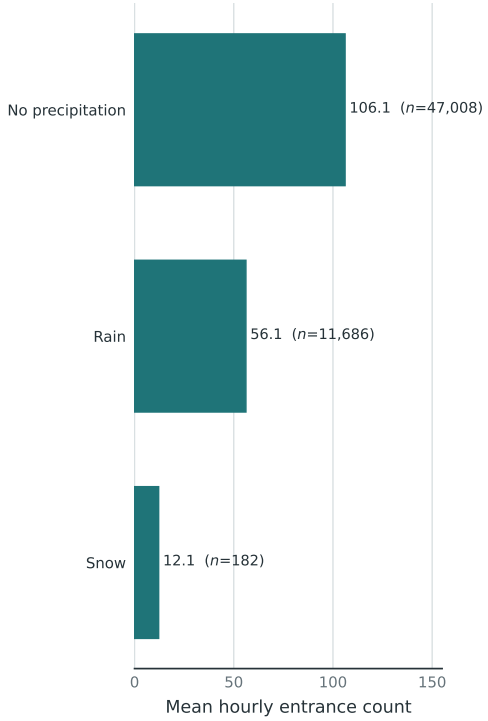


Figure 2. Mean hourly entrance count under different weather conditions. Labels show the number of hourly records. *n* means the number of samples.

2.4 Exploratory Data Analysis

We analyze the relationship between entrance count and two covariates, respectively.

Figure 2 shows mean hourly entrance count under different weather conditions. We can observe that mean hourly entrance count is generally higher when there is no precipitation and lower during rain or snow. However, these results describe associations rather than causal effects, and the estimates for snow conditions should be interpreted cautiously because of the small number of samples.

Figure 3 shows mean hourly entrance count on working days and on non-working days (weekends and public holidays) . A record is classified as a working day record only when its date falls on a weekday (Monday through Friday) and is not a public holiday. Other records are classified as non-working day records. For each category of records, the mean entrance count is calculated separately for every hour of the day. We observe that both categories show very low entrance count during the night, followed by a rapid increase after 8:00. Entrance count is substantially higher on non-working days between 9:00 and 16:00. Both categories reach the highest mean hourly entrance count at 14:00, with mean hourly entrance count of 148 on working days and mean hourly entrance count of 913 on non-working days. Both categories have slight downfall in mean hourly entrance count at 12:00.

3. FORECASTING MODEL

3.1 Feature Engineering

The feature set consists of entrance count, calendar variables, and weather condition. Calendar variables include hour of

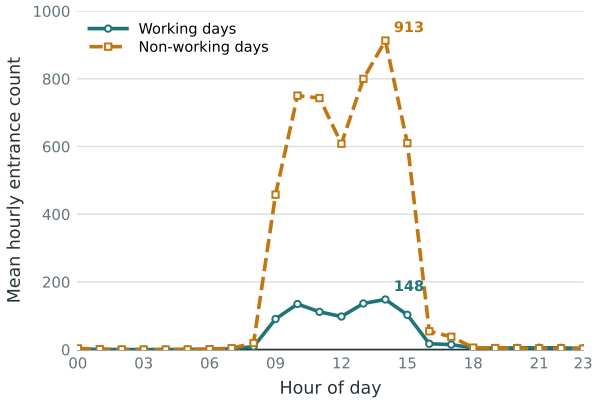


Figure 3. Mean hourly entrance count on working days and on non-working days (weekends and public holidays) .

day, weekday, and holiday information. Weather condition and calender variable are used as covariates that can influence entrance count.

Hour of day is encoded using sine and cosine transformations rather than as an integer from 0 to 23. This cyclical encoding preserves the relationship between the end and the beginning of a day. For example, 23:00 and 00:00 are only one hour apart, although their integer values appear far apart. The two encoded features are calculated as follows:

$$h_{\sin} = \sin\left(\frac{2\pi h}{24}\right), \qquad h_{\cos} = \cos\left(\frac{2\pi h}{24}\right), \tag{1}$$

where h denotes the hour of day. Weather condition is represented using one-hot encoding, as it is a categorical variable without a meaningful numerical order. Public holiday information is represented as a binary variable, with 1 indicating a non-working day and 0 indicating a working day. These transformations convert the original information into numerical features suitable for the GRU model.

3.2 Why choosing GRU?

GRU[6] is designed for sequential data and can learn how information from earlier timestamps relates to later outcomes. Hourly entrance counts form a sequential prediction problem in which earlier records may present information about later entrance counts. Compared with standard recurrent neural network, GRU uses reset and update gates to preserve useful information and reduce the loss of earlier temporal signals. This is appropriate for learning dependencies in both 24-hour entrance count sequence and covariate sequence.

Long Short-Term Memory (LSTM)[7] can provide similar sequence-modeling ability but uses more gates and a separate cell state. GRU requires fewer parameters and less computation. Given the moderate sequence lengths and the use of two separate recurrent encoders, GRU is chosen as a practical balance between modeling capacity and complexity.

[6] https://en.wikipedia.org/wiki/Gated_recurrent_unit
[7] https://en.wikipedia.org/wiki/Long_short-term_memory

Table 2. Performance of the model on the test set.

Metric	MAE	RMSE	R^2	SMAPE
Test result	65.40	157.79	0.854	97.39%

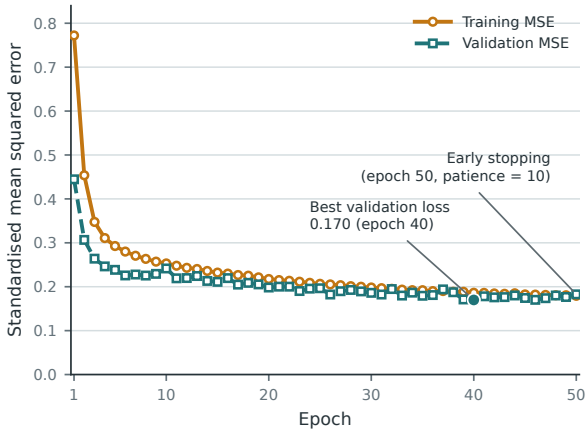


Figure 4. Training and validation loss curves of the model. Training ends at epoch 50 due to early stopping. The model parameters from epoch 40, which achieve the lowest validation loss, are retained for evaluation.

3.3 Model Development

The model receives two input sequences. The first has a shape of 24×1 and contains the entrance counts from the previous 24 hours. The second has a shape of 24×13 and contains the calendar and weather covariates from the next 24-hour period. The 13 covariates include two cyclical hour-of-day features, seven weekday indicators, one public-holiday indicator, and three weather-category indicators.

Each input sequence is processed by a separate single-layer GRU with 32 hidden units. The final hidden states of the two GRUs are concatenated into a 64-dimensional representation. A fully connected predictor then maps this representation to 24 outputs, producing entrance count predictions for the next 24 hours simultaneously.

The model is trained using Mean Squared Error (MSE) as the loss function, the Adam optimizer, and a batch size of 256. MSE measures the average absolute difference between the predicted and observed entrance count. It can be calculated as follow:

$$\text{MSE} = \frac{1}{NH}\sum_{i=1}^{N}\sum_{h=1}^{H}\left(y_{i,h} - \hat{y}_{i,h}\right)^2, \qquad H = 24, \tag{2}$$

where N is the number of samples, H is the forecast horizon, and $y_{i,h}$ and $\hat{y}_{i,h}$ denote the observed and predicted entrance counts for forecast hour h of sample i, respectively. Validation loss is monitored after every epoch. Early stopping with a patience of 10 epochs is applied (i.e, training is terminated when the validation loss fails to improve for 10 consecutive epochs) . The model parameters corresponding to the lowest validation loss are restored for final evaluation.

Table 3. Results of the covariate ablation experiment.

Configuration	MAE	RMSE	SMAPE
Full model	65.40	157.79	97.39%
Without weather	65.51	157.63	95.26%
Without calendar	88.04	197.92	104.98%

4. MODEL EVALUATION

4.1 Experimental Settings

The model is evaluated by four metrics, including MAE, RMSE, R^2, and SMAPE. MAE represents the average absolute prediction error, while RMSE gives greater weight to large errors. R^2 measures the proportion of variation in entrance count explained by the model. SMAPE measures the average percentage error relative to the magnitudes of the actual and predicted values.

A total of 48,124 samples are divided chronologically into training, validation, and test sets using an 80:10:10 ratio. The first 38,499 samples are used for training, the following 4,812 samples for validation, and the final 4,813 samples for testing. This chronological division ensures that the test samples occur later in time than the training and validation samples.

4.2 Main Results

Figure 4 show the training and validation loss curves of the model. We observe that training and validation losses decrease rapidly during the first few epochs. Validation loss reaches its minimum of 0.170 at epoch 40. Afterward, training loss continues to decrease while validation loss no longer improves, suggesting overfitting.

Table 2 shows the performance of the model on test set. We observe that the model achieves an MAE of 65.40 and an RMSE of 157.79. An R^2 of 0.854 shows that the model explains 85.4% of the variation in entrance count. Its SMAPE is 97.39%. Since the entrance counts of night hours are usually zero and SMAPE becomes unstable, it should be interpreted together with MAE, RMSE, and R^2.

4.3 Covariate Ablation Study

The study is conducted to examine the contributions of calendar and weather variables. Table 3 shows the results, and we observe that calendar variable contributes useful forecasting information. However, weather variable does not lead to significant improvement in forecast. The predictive improvement of weather covariates is concentrated mainly during precipitation periods, where they reduce the test set MAE from 67.00 to 51.09. However, this improvement is offset by poorer performance during the much more frequent non-precipitation periods and diluted by numerous closed-park hours with zero entrance counts.

5. CONCLUSIONS AND FUTURE WORK

This project develops a forecasting model for hourly entrance count at Liangzhu Archaeological Ruins Park. Raw records from June 2019 to June 2026 are aggregated into an hourly time series and combined with calendar and weather variables. Missing records between 1:00 and 6:00 are completed with zero entrance counts because these hours usually have no visitors. After continuity checks, 48,124 valid samples are generated. Each sample contains the entrance counts from the previous 24 hours, the calendar and weather covariates from the next 24-hour period, and the entrance counts from the next 24 hours. Exploratory analysis identifies differences in mean entrance count across weather conditions and between working and non-working days. A dual-GRU model separately encodes the historical entrance count sequence and 24-hour covariate sequence, which achieves an MAE of 65.40, an RMSE of 157.79, an R^2 of 0.854, and an SMAPE of 97.39% on the test set.

For future work, we will use the entrance counts from longer historical period and longer weather forecasts to predict entrance counts over a longer period of time. To be more consistent with practical conditions, we will use historical weather forcasts as weather variable, rather than using actual weather conditions. Finally, we will develop LSTM based model and compare its performance with that of GRU based model.